CariSurg MedTech Pathways Research Project
Preliminary Proposal

Project Title:
Explainable AI-Assisted Emergency Department Triage Support System for Resource-Constrained Healthcare Settings

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Statement of the Problem

Emergency department triage frequently depends on subjective clinical assessment, resulting in inconsistent patient prioritization and avoidable under-triage of high-acuity cases during periods of overcrowding. Recent literature demonstrates that AI-assisted triage systems can improve prediction accuracy and decision consistency; however, most studies rely on retrospective, single-site datasets with limited explainability and no reported prospective validation in Caribbean or small-island healthcare contexts [1–7]. This project proposes evaluating an explainable machine learning decision-support approach to improve triage consistency while maintaining clinician oversight.

Previous Work

[1] This review examined how artificial intelligence is being applied to emergency department triage to reduce overcrowding and improve patient prioritization. AI was primarily used through machine learning and natural language processing approaches that processed patient data to support triage decisions and optimize resource allocation. The review reported improvements in wait times, reductions in time-to-treatment, and more efficient bed and staffing allocation. However, the authors identified challenges including algorithmic bias, clinician distrust, and concerns surrounding black-box decision-making.

[2] This scoping review explored the use of artificial intelligence across emergency department triage systems to improve patient classification and outcome prediction. AI was applied using machine learning algorithms trained on patient demographics, symptoms, and clinical indicators to support prioritization decisions. The review found that AI systems frequently outperformed traditional triage approaches, with some models demonstrating substantially improved predictive performance. The authors highlighted limitations related to retrospective datasets, inconsistent preprocessing methods, and limited external validation.

[3] This study developed TriageIntelli, an AI-assisted triage system designed to support emergency assessment and reduce delays caused by increasing patient volumes. Artificial intelligence was implemented through six supervised machine learning models trained on emergency department data and combined into a stacked ensemble architecture. The final stacked model achieved the highest performance, reaching 80.05% classification accuracy

and an AUROC of 0.93. Despite strong performance, the study was limited by its reliance on historical data collected from only two Korean emergency departments.

[4] This study investigated whether machine learning and natural language processing could improve emergency department triage by predicting patient clinical disposition. AI was used to analyze structured triage data together with free-text clinical notes to identify patterns associated with admission outcomes. The results demonstrated improved predictive capability compared with conventional approaches and highlighted the value of combining structured and unstructured clinical information. Limitations included challenges associated with model generalization and dependence on local documentation practices.

[5] This comparative study evaluated the performance of large language models in emergency medicine triage tasks relative to inexperienced human decision-makers. AI was implemented using large language models to generate triage recommendations across simulated emergency scenarios. The findings demonstrated promising performance and highlighted the potential of language models to support triage workflows and decision-making. However, concerns remained regarding reliability, transparency, and the safe deployment of generative AI systems in clinical environments.

[6]This study developed and validated a machine learning model to predict 72 hour emergency department return visit admissions (RVA) using electronic health record data from multiple urban hospitals. Several algorithms, including logistic regression, gradient boosting methods, and a clustering-based approach, were evaluated to identify high-risk patients. The final model achieved strong predictive performance in the development dataset (AUC up to 0.87) but showed reduced performance in external validation (AUC 0.75), highlighting challenges in generalizability across clinical settings. The study also incorporated interpretability techniques such as SHAP analysis and clinician chart review to assess model reasoning and clinical relevance. While the model demonstrated potential for identifying patients at risk of deterioration after discharge, the authors emphasized that limited clinical actionability, dataset imbalance, and variation in hospital workflows restrict real-world implementation without further refinement and external validation.

[7] This study developed an interpretable machine learning emergency department triage model designed to improve prediction performance while addressing class imbalance in emergency datasets. Artificial intelligence was implemented through an interpretable ML framework that generated transparent triage recommendations and prioritization scores rather than relying on black-box outputs. The study reported improved triage performance and demonstrated that explainable approaches may strengthen clinician trust and adoption. Limitations included dependence on local population characteristics and uncertainty regarding transferability across different emergency healthcare settings.

Proposed Solution

This project proposes an explainable AI-assisted emergency department triage decision-support system for resource-constrained healthcare settings. The system will use supervised machine learning models trained on structured clinical data such as vital signs, demographics, and presenting complaints to predict patient acuity levels. A baseline logistic regression model will be compared with ensemble methods such as gradient boosting to identify the most effective approach. Outputs will include a triage category (low, moderate, high, or critical) and a confidence score.

To address limitations of “black-box” models identified in prior studies, the system will integrate SHAP-based explainability to show how individual features influence each prediction, improving transparency and clinician trust. The tool will function as a clinician-in-the-loop decision-support system rather than replacing clinical judgment. Evaluation will use retrospective data and metrics including AUC, F1-score, and sensitivity, with emphasis on reducing under-triage. The study will also assess feasibility for deployment in Caribbean healthcare environments

References

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